

Estimating smooth structural change in cointegrated system with local-to-unity regressors

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Abstract

In this note, we derive the asymptotic properties of the Nadaraya-Watson type local level regression estimator of time-varying cointegrating coefficients with local-to-unity regressors. It is shown that weighted signal matrix becomes asymptotically singular and a rotation decomposition is developed to restore the limit theory. The theoretical findings are illustrated via a Monte Carlo study.

Keywords: Time-varying cointegration, Local-to-unity, Local level regression

JEL classification: C13, C14, C32

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1. Introduction

There has been growing interest in the limit theory of nonparametric time-varying coefficient cointegration regressions. This literature dates back to Park and Hahn (1999), who develop a series estimator for time-varying cointegrating vectors in a single-equation framework. Within a kernel estimation setting, Phillips et al. (2017) show that conventional asymptotic theory under stationarity breaks down when regressors are integrated in a single equation framework, due to a kernel-induced degeneracy in the weighted signal matrix. They introduce a novel rotational decomposition to resolve this issue and establish the limit theory. Building on this work, Li et al. (2020) extend the previous work to allow for cointegrated regressors and mixtures of stochastic and deterministic trend components, while Kapetanios et al. (2020) develop a nonparametric test for time-varying cointegration. A common feature of the aforementioned literature is that regressors are exactly integrated, with autoregressive roots fixed at unity. However, a large body of empirical evidence suggests that economic and financial time series often exhibit strong persistence without being precisely unit root processes, with autoregressive roots lying in a neighborhood of unity.

In this note, we establish limit theory for Nadaraya-Watson local level estimation of time-varying cointegrating coefficients in a cointegrated system with local-to-unity regressors. We show that conventional kernel methods encounter a degeneracy problem due to asymptotic singularity of the weighted signal matrix. To address this feature, we employ a rotational decomposition to develop the limit theory. The resulting asymptotics exhibit two distinct convergence rates in the different directions associated with the rotations: a *type I* super-consistency rate of $n\sqrt{h}$ and a *type II* super-consistency rate of nh , where $h := h_n$ denotes the bandwidth parameter. A Monte Carlo study is conducted to assess the finite-sample performance of the estimator.

The rest of the note is organised as follows. Section 2 introduces the model, lays out the assumptions, and presents the estimator. Section 3 develops the asymptotic theory. Section 4 presents a Monte Carlo study and Section 5 concludes. The Appendix presents two Lemmas needed to establish the main theoretical results. All mathematical proofs, auxiliary lemmas, and additional simulation results are collected in the Online Appendix.

Notation: *a.s.* means almost surely. $\|\cdot\|$ is the Euclidean norm. $\Rightarrow$ signifies weak convergence. $\mathbf{K}_C(\cdot)$ denotes a vector Ornstein–Uhlenbeck process initialized at $\mathbf{K}_C(0) = \mathbf{0}_{k \times 1}$ and satisfying the following stochastic differential equation $d\mathbf{K}_C(r) = \mathbf{C}\mathbf{K}_C(r)dr + d\mathbf{B}(r)$, where $\mathbf{B}(r)$ is a vector Brownian motion. All weak convergence statements are understood to hold in $D([0, 1])$ equipped with the Skorokhod metric.

2. Model and the estimator

We consider the triangular system (cf. Phillips (1991)) with time-varying coefficients

$$\mathbf{y}_t = \mathbf{A}_t \mathbf{X}_t + \mathbf{u}_{0t}, \quad (1)$$

$$\mathbf{X}_t = \mathbf{R}_n \mathbf{X}_{t-1} + \mathbf{u}_{xt}, \quad (2)$$

$$\mathbf{R}_n = \mathbf{I}_k + \frac{\mathbf{C}}{n}, \quad \mathbf{C} = \text{diag}(c_1, \dots, c_k), \quad (3)$$

for $t = 1, 2, \dots, n$. The system is initialized at some $\mathbf{X}_0 = O_p(1)$. $\mathbf{A}_t := \mathbf{A}(t/n)$ is an $d \times k$ matrix of time-varying "cointegrating" coefficients, with elements measured as functions of scaled time point t/n . c_i is assumed to be negative for all i , where $i = 1, 2, \dots, k$. When $\mathbf{A}_t = \mathbf{A}_0$ for all t , the model reduces to the near-integrated regression framework studied in Phillips (1988).

For any fixed $z_0 \in (0, 1)$, under a suitable smoothness condition on $\mathbf{A}(\cdot)$ (Assumption 2), we have

$$\mathbf{A}(t/n) = \mathbf{A}(z_0) + O\left(\left|\frac{t}{n} - z_0\right|\right) \approx \mathbf{A}(z_0), \quad (4)$$

where t/n lies in a small neighbourhood of z_0 . The Nadaraya-Watson type local level regression estimator of $\mathbf{A}(z_0)$ has the usual form given by²

$$\widehat{\mathbf{A}}_n(z_0) = \left[\sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \mathbf{y}_t \mathbf{X}_t^\top \right] \left[\sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^\top \right]^+, \quad (5)$$

where $K(\cdot)$ is some kernel function, and h is a bandwidth parameter. We make the following assumptions on the kernel function $K(\cdot)$ and bandwidth h .

Assumption 1. (i) The kernel function $K(\cdot)$ is Lipschitz-continuous, positive, symmetric, and has compact support $[-1, 1]$ with $\int_{-1}^1 K(u) du = 1$.

(ii) The bandwidth $h := h_n$ satisfies $h \rightarrow 0$ and $nh \rightarrow \infty$, as $n \rightarrow \infty$.

The smoothness condition on $\mathbf{A}(\cdot)$ is given below.

Assumption 2. Let $a^{(i_1, i_2)}(t/n)$ be the (i_1, i_2) th elements in $\mathbf{A}(t/n)$, where $i_1 = 1, 2, \dots, d$, $i_2 = 1, 2, \dots, k$. $\mathbf{A}^{(i_1, i_2)}(\cdot)$ is a real function and is continuously differentiable on $(0, 1)$.

²Throughout the note, matrices involving the inverse of the weighted signal matrix are written using the Moore-Penrose generalized inverse $(\cdot)^+$ rather than the usual inverse $(\cdot)^{-1}$, to accommodate the possible degeneracy of the limit matrix. Once nonsingularity is established *a.s.*, the generalized inverse coincides with the ordinary inverse, and the distinction between $(\cdot)^+$ and $(\cdot)^{-1}$ becomes immaterial.

We assume that the innovations $\mathbf{u}_t = (\mathbf{u}_{0t}^\top, \mathbf{u}_{xt}^\top)^\top$ are determined according to the linear process

$$\mathbf{u}_t = \Phi(\mathcal{L})\boldsymbol{\varepsilon}_t = \sum_{j=0}^{\infty} \Phi_j \boldsymbol{\varepsilon}_{t-j}, \quad (6)$$

where $\Phi(\mathcal{L}) = \sum_{j=0}^{\infty} \Phi_j \mathcal{L}^j$, Φ_j is a sequence of $(d+k) \times (d+k)$ matrices, $\mathcal{L}$ is the lag operator, and $(\boldsymbol{\varepsilon}_t)_t$ is a sequence of independent and identically distributed (i.i.d.) random vectors with dimension $d+k$. We impose the following conditions.

Assumption 3. (i) Let $(\boldsymbol{\varepsilon}_t)_t$ be i.i.d. $(d+k)$ -dimensional random vectors with $\mathbb{E}(\boldsymbol{\varepsilon}_t) = \mathbf{0}$, $\Sigma_\varepsilon := \mathbb{E}(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top) > 0$, and $\mathbb{E}\|\boldsymbol{\varepsilon}_t\|^p < \infty$, for some $p > 4$. (ii) The coefficient matrices in (6) satisfy $\sum_{j=0}^{\infty} j\|\Phi_j\| < \infty$ and $\Phi(1) = \sum_{j=0}^{\infty} \Phi_j$ has full row rank.

Under Assumption 3, $\mathbf{u}_t$ has covariance matrix $\Sigma = \sum_{j=0}^{\infty} \Phi_j \Sigma_\varepsilon \Phi_j^\top$, with $\mathbb{E}\|\mathbf{u}_t\|^p < \infty$, for some $p > 4$. By functional central limit theory (FCLT) for linear process (cf. Phillips and Solo (1992)), we have, for $0 < r \leq 1$

$$n^{-1/2} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_t \Rightarrow \mathbf{B}(r), \quad (7)$$

with $\mathbf{B}(r)$ being $(d+k)$ -dimensional Brownian motion with full covariance matrix $\Omega = \Phi(1) \Sigma_\varepsilon \Phi(1)^\top$. For the ease of theoretical development, we partition Σ , $\Phi(1)$, and Ω comfortably with $\mathbf{u}_t$ as

$$\Sigma = \begin{bmatrix} \Sigma_{00} & \Sigma_{0x} \\ \Sigma_{x0} & \Sigma_{xx} \end{bmatrix}, \quad \Phi(1) = \begin{bmatrix} \Phi_0(1) \\ \Phi_x(1) \end{bmatrix}, \quad \Omega = \begin{bmatrix} \Omega_{00} & \Omega_{0x} \\ \Omega_{x0} & \Omega_{xx} \end{bmatrix},$$

where $\Phi_0(1)$ and $\Phi_x(1)$ are of dimension $d \times (d+k)$ and $k \times (d+k)$, respectively. Using the Beveridge–Nelson (BN) decomposition, we have the following decomposition for $\mathbf{u}_t$:

$$\mathbf{u}_t = \Phi(1)\boldsymbol{\varepsilon}_t - \Delta \widetilde{\mathbf{u}}_t, \quad \text{for } \widetilde{\mathbf{u}}_t = \sum_{j=0}^{\infty} \widetilde{\Phi}_j \boldsymbol{\varepsilon}_{t-j}, \quad \widetilde{\Phi}_j = \sum_{k=j+1}^{\infty} \Phi_k, \quad (8)$$

where . Corresponding to the partition of $\mathbf{u}_t$, we write

$$\mathbf{u}_{0t} = \sum_{j=0}^{\infty} \Phi_{0j} \boldsymbol{\varepsilon}_{t-j}, \quad \widetilde{\mathbf{u}}_{0t} = \sum_{j=0}^{\infty} \widetilde{\Phi}_{0j} \boldsymbol{\varepsilon}_{t-j}, \quad (9)$$

and also

$$\mathbf{u}_{xt} = \sum_{j=0}^{\infty} \Phi_{xj} \boldsymbol{\varepsilon}_{t-j}, \quad \widetilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty} \widetilde{\Phi}_{xj} \boldsymbol{\varepsilon}_{t-j}, \quad (10)$$

where Φ_{0j} , $\widetilde{\Phi}_{0j}$ are of dimension $d \times (d+k)$ and Φ_{xj} , $\widetilde{\Phi}_{xj}$ are of dimension $K \times (d+k)$, respectively.

In addition, the limit theory also involves the partitioned components of the one-sided long-run covariance matrix

$$\mathbf{\Lambda} := \begin{bmatrix} \mathbf{\Lambda}_{00} & \mathbf{\Lambda}_{0x} \\ \mathbf{\Lambda}_{x0} & \mathbf{\Lambda}_{xx} \end{bmatrix} = \sum_{j=1}^{\infty} \mathbb{E}(\mathbf{u}_t \mathbf{u}_{t-j}^{\top}).$$

As usual, we have $\mathbf{\Omega} = \mathbf{\Sigma} + \mathbf{\Lambda} + \mathbf{\Lambda}^{\top}$ and

$$\mathbf{\Lambda} = \sum_{j=1}^{\infty} \sum_{k=0}^{\infty} \mathbf{\Phi}_{k+j} \mathbf{\Sigma}_{\varepsilon} \mathbf{\Phi}_k^{\top} = \sum_{k=0}^{\infty} \left(\sum_{j=1}^{\infty} \mathbf{\Phi}_{k+j} \right) \mathbf{\Sigma}_{\varepsilon} \mathbf{\Phi}_k^{\top} = \sum_{k=0}^{\infty} \tilde{\mathbf{\Phi}}_k \mathbf{\Sigma}_{\varepsilon} \mathbf{\Phi}_k^{\top} = \mathbb{E}(\tilde{\mathbf{u}}_t \mathbf{u}_t^{\top}).$$

3. Asymptotic theory

In this section, we derive the asymptotic properties of the local level regression estimator (5). We shall first consider the asymptotic behavior of the weighted signal matrix $\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^{\top}$. This is characterised in the next lemma.

Lemma 1. *If Assumptions 1-3 are satisfied. Then, for any fixed $z_0 \in (0, 1)$,*

$$\frac{1}{n^2 h} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^{\top} \Rightarrow \mathbf{K}_C(z_0) \mathbf{K}_C^{\top}(z_0).$$

Observe that $\mathbf{K}_C(z_0)$ is a Gaussian process which, for fixed $z_0 \in (0, 1)$, has the distribution

$$\mathbf{K}_C(z_0) \equiv \mathcal{N}(\mathbf{0}_{k \times 1}, \mathbf{\Sigma}_K(z_0)),$$

where $\mathbf{\Sigma}_K(z_0) := \int_0^{z_0} e^{(z_0-s)\mathbf{C}} \mathbf{\Omega}_{xx} e^{(z_0-s)\mathbf{C}} ds$. Consequently, $\mathbf{K}_C(z_0) \mathbf{K}_C^{\top}(z_0)$ is a (possibly singular) Wishart variate with one degree of freedom and scale matrix $\mathbf{\Sigma}_K(z_0)$, and hence has rank one almost surely whenever $\mathbf{\Sigma}_K(z_0)$ is nonsingular. Therefore, the limiting matrix in Lemma 1 is rank one, implying that $\frac{1}{n^2 h} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^{\top}$ is asymptotically singular whenever $\dim(\mathbf{X}_t) = k > 1$.

To deal with this limiting degeneracy issue, we follow the rotation decomposition approach introduced in Phillips et al. (2017) for the exactly integrated case. Define $\mathbf{k} \equiv \mathbf{k}_{z_0} = \mathbf{K}_C(z_0)$ and set

$$\mathbf{q} = \frac{\mathbf{k}}{(\mathbf{k}^{\top} \mathbf{k})^{1/2}} = \frac{\mathbf{k}}{\|\mathbf{k}\|}.$$

Let $\mathbf{q}_n^{\perp}$ be the $k \times (k-1)$ be the orthogonal component matrix such that $\mathbf{Q} = (\mathbf{q}, \mathbf{q}^{\perp})$, $\mathbf{Q}^{\top} \mathbf{Q} = \mathbf{I}_k$. The sample version of these quantities is

$$\mathbf{q}_n = \frac{\mathbf{k}_n}{(\mathbf{k}_n^{\top} \mathbf{k}_n)^{1/2}} = \frac{\mathbf{k}_n}{\|\mathbf{k}_n\|}, \quad \mathbf{k}_n \equiv \mathbf{k}_n(z_0) = \frac{1}{\sqrt{n}} \mathbf{X}_{z_n}.$$

Moreover, let $\mathbf{Q}_n = (\mathbf{q}_n, \mathbf{q}_n^\perp)$, $\mathbf{Q}_n^\top \mathbf{Q}_n = \mathbf{I}_k$ and defined the standardization matrix

$$\mathbf{D}_n = \text{diag} \left\{ n \sqrt{h}, (nh) \mathbf{I}_{k-1} \right\}.$$

Note that both $\mathbf{Q}$ and $\mathbf{Q}_n$ are random, path dependent, and localized to the point z_0 and $z_n = \lfloor (z_0 - h)n \rfloor$, respectively.

We have the following decomposition

$$\begin{aligned} & (\hat{\mathbf{A}}_n(z_0) - \mathbf{A}(z_0)) \mathbf{Q}_n \mathbf{D}_n \\ &= \left[\left(\sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) (\mathbf{A}_t - \mathbf{A}(z_0)) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right] \left[\mathbf{D}_n^+ \mathbf{Q}_n^\top \left(\sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right]^+ \\ &+ \left[\left(\sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) \mathbf{u}_{0t} \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right] \left[\mathbf{D}_n^+ \mathbf{Q}_n^\top \left(\sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right]^+ \\ &:= \mathbf{B}_n(z_0) \Delta_n^+(z_0) + \mathbf{\Gamma}_n(z_0) \Delta_n^+(z_0). \end{aligned} \quad (11)$$

The following theorem gives the asymptotic distribution of $\hat{\mathbf{A}}_n(z_0)$.

Theorem 1. *If Assumptions 1-3 are satisfied and $nh^{3/2} \rightarrow 0$. For any fixed $z_0 \in (0, 1)$, we have*

$$(\hat{\mathbf{A}}_n(z_0) - \mathbf{A}(z_0)) \mathbf{Q}_n \mathbf{D}_n \Rightarrow \mathbf{\Gamma}(z_0) \Delta^+(z_0). \quad (12)$$

$\Delta(z_0)$ is defined as

$$\Delta(z_0) := \begin{bmatrix} \Delta(z_0, 1) & \Delta(z_0, 2) \\ \Delta(z_0, 2)^\top & \Delta(z_0, 3) \end{bmatrix}$$

with

$$\begin{aligned} \Delta(z_0, 1) &= \mathbf{k}^\top \mathbf{k}, \\ \Delta(z_0, 2) &= \sqrt{2} \|\mathbf{k}\| \left[\int_{-1}^1 K(r) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) dr \right] \mathbf{q}^\perp, \\ \Delta(z_0, 3) &= 2 (\mathbf{q}^\perp)^\top \left[\int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) dr \right] \mathbf{q}^\perp, \end{aligned}$$

and is nonsingular a.s.. $\mathbf{\Gamma}(z_0)$ is defined as

$$\mathbf{\Gamma}(z_0) = \begin{bmatrix} \underbrace{\mathbf{\Gamma}(z_0, 1)}_{d \times 1} & \underbrace{\mathbf{\Gamma}(z_0, 2)}_{d \times (k-1)} \end{bmatrix},$$

with

$$\begin{aligned}\Gamma(z_0, 1) &= \sqrt{2} \|\mathbf{k}\| \int_{-1}^1 K(r) d\mathbf{B}_0^* \left(\frac{r+1}{2} \right), \\ \Gamma(z_0, 2) &= 2 \left[\int_{-1}^1 K(r) d\mathbf{B}_0^* \left(\frac{r+1}{2} \right) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) + \frac{1}{2} (\boldsymbol{\Sigma}_{0x} + \boldsymbol{\Lambda}_{0x}) \right] \mathbf{q}^\perp.\end{aligned}$$

and $(\mathbf{K}_C^{*\top}(r), \mathbf{B}_0^{*\top}(r))^\top$ is an independent copy of $(\mathbf{K}_C^\top(r), \mathbf{B}_0^\top(r))^\top$. $\mathbf{B}_0(r)$ is a k -vector Brownian motion with covariance matrix $\boldsymbol{\Omega}_{00}$.

Several comments are in order. First, as we expect, two types of convergence rates are involved. In the direction of $\mathbf{q}_n$, we have a faster convergence rate given by

$$(\widehat{\mathbf{A}}_n(z_0) - \mathbf{A}(z_0)) \mathbf{q}_n = O_p \left(\frac{1}{n \sqrt{h}} \right),$$

which is called *type I* super-consistency as in Phillips et al. (2017). In the direction of $\mathbf{q}_n^\perp$, (12) gives

$$(\widehat{\mathbf{A}}_n(z_0) - \mathbf{A}(z_0)) \mathbf{q}_n^\perp = O_p \left(\frac{1}{nh} \right),$$

which is slower than the $n \sqrt{h}$ rate in the direction of $\mathbf{q}_n$. We also name it *type II* super-consistency.

Second, while in the direction of $\mathbf{q}_n$, $\int_{-1}^1 K(r) d\mathbf{B}_0^* \left(\frac{r+1}{2} \right)$ is distributed as $\mathcal{N} \left(0, \frac{1}{2} v_0 \boldsymbol{\Omega}_{00} \right)$ with $v_0 = \int_{-1}^1 K^2(u) du$. This is identical to the unit root case considered in Phillips et al. (2017). In the direction of $\mathbf{q}_n^\perp$, $\int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) d\mathbf{B}_0^* \left(\frac{r+1}{2} \right)$ has a nonstandard limiting distribution, which is also different from the unit root case.

Third, the bandwidth condition $nh^{3/2} \rightarrow 0$ ensures that the smoothing bias is asymptotically negligible. As demonstrated in the proof of Theorem 1, in contrast to the stationary case, the bias process is stochastic and path dependent.

4. Monte Carlo study

We use the Data Generating Process (D.G.P.) as in Phillips et al. (2017):

$$y_t = \mathbf{A}_t^\top \mathbf{X}_t + u_{0t},$$

where $\mathbf{A}_t = (a_{1t}, a_{2t})^\top$ has the following functional forms

- M1: $a_{1t} = 1 + \frac{t}{n}$, $a_{2t} = e^{-\frac{t}{n}}$;
- M2: $a_{1t} = \cos(2\pi t/n)$, $a_{2t} = \sin(2\pi t/n)$.

The regressor $\mathbf{X}_t = (x_{1t}, x_{2t})^\top$ is generated according to $x_{i,t} = \left(1 - \frac{1}{n}\right)x_{i,t-1} + u_{i,xt}$ for $i = 1, 2$. We generate $\mathbf{u}_t = (u_{0t}, u_{1,xt}, u_{2,xt})^\top$ from a VAR(1) process

$$\mathbf{u}_t = \mathbf{\Pi}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{\Pi} = \text{diag}(\rho, \rho_1, \rho_2)$ and the innovation vector $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}_{3 \times 1}, (1 - \lambda)\mathbf{I}_3 + \lambda\mathbf{J}_3)$. We consider six cases as in Phillips et al. (2017):

- Case 1: $\rho = \rho_1 = \rho_2 = 0, \lambda = 0, (z_0, h) = \left(\frac{1}{4}, \frac{1}{6}\right)$;
- Case 2: $\rho = \rho_1 = \rho_2 = 0, \lambda = 0, (z_0, h) = \left(\frac{1}{2}, \frac{1}{3}\right)$;
- Case 3: $\rho = \rho_1 = \rho_2 = 0, \lambda = 0, (z_0, h) = \left(\frac{3}{4}, \frac{1}{2}\right)$;
- Case 4: $\rho = 0.5, \rho_1 = -0.5, \rho_2 = 0.5, \lambda = 0.5, (z_0, h) = \left(\frac{1}{4}, \frac{1}{6}\right)$;
- Case 5: $\rho = 0.5, \rho_1 = -0.5, \rho_2 = 0.5, \lambda = 0.5, (z_0, h) = \left(\frac{1}{2}, \frac{1}{3}\right)$;
- Case 6: $\rho = 0.5, \rho_1 = -0.5, \rho_2 = 0.5, \lambda = 0.5, (z_0, h) = \left(\frac{3}{4}, \frac{1}{2}\right)$.

To compute the estimator (5), we employ the uniform kernel, $K(u) = \frac{1}{2}\mathbf{1}(|u| \leq 1)$. We consider two sample sizes: $n = 500, 1000$. The Monte Carlo analysis is based on $R = 2,000$ replications.

As in Section 3, we use the following notations: $z_n = \lfloor (z_0 - h)n \rfloor$, $\mathbf{X}_{z_n} = (x_{1,z_n}, x_{2,z_n})^\top$, $\mathbf{k}_n \equiv \mathbf{k}_n(z_0) = \frac{1}{\sqrt{n}}\mathbf{X}_{z_n}$, $\mathbf{q}_n(z_0) = \frac{\mathbf{k}_n}{\|\mathbf{k}_n\|} = \left(\frac{x_{1,z_n}}{\|\mathbf{X}_{z_n}\|}, \frac{x_{2,z_n}}{\|\mathbf{X}_{z_n}\|}\right)^\top \equiv (q_{1n}(z_0), q_{2n}(z_0))^\top$, and $\mathbf{q}_n^\perp(z_0) = (p_{1n}(z_0), p_{2n}(z_0))^\top$. Given that $\mathbf{Q}_n(z_0) = [\mathbf{q}_n(z_0), \mathbf{q}_n^\perp(z_0)]$ has to be orthonormal, we set $p_{1n}(z_0) = q_{2n}(z_0)$ and $p_{2n}(z_0) = -q_{1n}(z_0)$. The performance of the estimators is evaluated based on the transformed quantities

$$\begin{aligned}\tilde{a}_{1n}(z_0) &= q_{1n}(z_0) [\hat{a}_{1n}(z_0) - a_1(z_0)] + q_{2n}(z_0) [\hat{a}_{2n}(z_0) - a_2(z_0)], \\ \tilde{a}_{2n}(z_0) &= p_{1n}(z_0) [\hat{a}_{1n}(z_0) - a_1(z_0)] + p_{2n}(z_0) [\hat{a}_{2n}(z_0) - a_2(z_0)],\end{aligned}$$

where $\hat{\mathbf{A}}_t = (\hat{a}_{1n}(z_0), \hat{a}_{2n}(z_0))^\top$. We report $\frac{1}{R} \sum_{j=1}^R \tilde{a}_{in,j}(z_0)$ for $i = 1, 2$, where $\tilde{a}_{in,j}(z_0)$ is the value of $\tilde{a}_{in}(z_0)$ at the j th iteration.

The simulation results are reported in Table 1. In most of the cases, $|\tilde{a}_{1n}(z_0)|$ is smaller than $|\tilde{a}_{2n}(z_0)|$, which is broadly consistent with the asymptotic theory developed in Section 3 that $|\tilde{a}_{1n}(z_0)|$ converges to zero at a faster rate than $|\tilde{a}_{2n}(z_0)|$.

5. Conclusion

This paper establishes the limit theory for Nadaraya–Watson type local level estimation of time-varying cointegrating coefficients in systems with local-to-unity regressors. We show that, conventional kernel

Table 1: Small sample properties of the estimator (5) for near integrated regressors: absolute average of $\tilde{a}_m(z_0)$

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
M1	<i>n</i> = 500					
$ \tilde{a}_{1n}(z_0) $	0.060	0.101	0.155	0.057	0.075	0.113
$ \tilde{a}_{2n}(z_0) $	0.087	0.144	0.162	0.145	0.188	0.185
	<i>n</i> = 1000					
$ \tilde{a}_{1n}(z_0) $	0.060	0.099	0.158	0.052	0.079	0.108
$ \tilde{a}_{2n}(z_0) $	0.088	0.146	0.159	0.134	0.184	0.190
M2	<i>n</i> = 500					
$ \tilde{a}_{1n}(z_0) $	0.316	0.578	0.711	0.261	0.641	0.855
$ \tilde{a}_{2n}(z_0) $	0.413	0.680	0.690	0.484	1.553	1.113
	<i>n</i> = 1000					
$ \tilde{a}_{1n}(z_0) $	0.301	0.575	0.709	0.256	0.648	0.873
$ \tilde{a}_{2n}(z_0) $	0.424	0.682	0.675	0.500	1.569	1.115

methods encounter a degeneracy problem due to asymptotic singularity of the weighted signal matrix. We employ a rotational decomposition to develop the limit theory. Monte Carlo simulations are conducted illustrate the theoretical findings.

An important extension of the present analysis is to develop corresponding limit theory for mildly integrated and mildly explosive regressors with autoregressive roots of the form $\rho_{ni} = 1 + c_i/n^\alpha$, where $\alpha \in (0, 1)$ and $c_i \neq 0$ are constant parameters. We leave this extension for future research.

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Appendix A. Technical Lemmas

The appendix presents two lemmas required for the proof of Theorem 1. Detailed proofs of both lemmas are provided in Section A of the Online Supplement.

Lemma A1. *If Assumptions 1-3 are satisfied. Then, for any fixed $z_0 \in (0, 1)$,*

$$\Delta_n(z_0) \Rightarrow \Delta(z_0),$$

where $\Delta_n(z_0)$ is defined as in (11) and $\Delta(z_0)$ is defined as in Theorem 1 and is nonsingular a.s..

Lemma A2. *If Assumptions 1-3 are satisfied. Then, for any fixed $z_0 \in (0, 1)$,*

$$\Gamma_n(z_0) \Rightarrow \Gamma(z_0),$$

where $\Gamma_n(z_0)$ is defined as in (11) and $\Gamma(z_0)$ is defined as in Theorem 1.

Online Appendix: Estimating smooth structural change in cointegrated system with local-to-unity regressors

The online appendix is organized as follows. Section A provides proofs of Lemma 1, Theorem 1, Lemma A1, and Lemma A2. Section B presents additional lemmas needed to establish the main theoretical results and their proofs. Section C reports some additional simulation results.

Notation: *a.s.* means almost surely. $\|\cdot\|$ is the Euclidean norm. $\lfloor \cdot \rfloor$ denotes the integer part. The operator $\xrightarrow{P}$ denotes convergence in probability. $\Rightarrow$ signifies weak convergence. $x_n = O(y_n)$ states that the deterministic sequence x_n is at most of order y_n . $x_n = O_p(y_n)$ states that the vector of random variables x_n is at most of order y_n in probability, and $x_n = o_p(y_n)$ is of smaller order in probability than y_n . $\mathbf{K}_C(\cdot)$ denotes a vector Ornstein–Uhlenbeck process initialized at $\mathbf{K}_C(0) = \mathbf{0}_{k \times 1}$ and satisfying the following stochastic differential equation $d\mathbf{K}_C(r) = \mathbf{C}\mathbf{K}_C(r)dr + d\mathbf{B}(r)$, where $\mathbf{B}(r)$ is a vector Brownian motion. All weak convergence statements are understood to hold in $D([0, 1])$ equipped with the Skorokhod metric.

A Mathematical proofs

A.1 Proof of Lemma 1

Take a neighborhood $N_{z_0}(h) = [\lfloor (z_0 - h)n \rfloor, \lfloor (z_0 + h)n \rfloor]$ of $\lfloor z_0 n \rfloor$ and let $z_n = \lfloor (z_0 - h)n \rfloor$. We have the following representation:

$$\begin{aligned} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^\top &= \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) (\mathbf{X}_t - \mathbf{X}_{z_n} + \mathbf{X}_{z_n}) (\mathbf{X}_t - \mathbf{X}_{z_n} + \mathbf{X}_{z_n})^\top \\ &= \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \mathbf{X}_{z_n} \mathbf{X}_{z_n}^\top + \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) (\mathbf{X}_t - \mathbf{X}_{z_n}) \mathbf{X}_{z_n}^\top \\ &\quad + \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \mathbf{X}_{z_n} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top + \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) (\mathbf{X}_t - \mathbf{X}_{z_n}) (\mathbf{X}_t - \mathbf{X}_{z_n})^\top. \end{aligned}$$

By Assumption 1 and a Riemann-sum approximation, we have

$$\frac{1}{nh} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \longrightarrow \int_{-1}^1 K(u) du = 1, \quad 0 < z_0 < 1.$$

Together with Lemma B2(i) and Slutsky's theorem, this implies

$$\frac{1}{n^2 h} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \mathbf{X}_{z_n} \mathbf{X}_{z_n}^\top = \left(\frac{\mathbf{X}_{z_n} \mathbf{X}_{z_n}^\top}{n} \right) \left(\frac{1}{nh} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \right) \Rightarrow \mathbf{K}_C(z_0) \mathbf{K}_C^\top(z_0). \quad (1)$$

Lemma B2 implies $\|\mathbf{X}_{z_n}\| = O_p(\sqrt{n})$ and, uniformly for $t \in \bar{N}_{nz_0}(h)$, $\|\mathbf{X}_t - \mathbf{X}_{z_n}\| = O_p(\sqrt{nh})$, where $\bar{N}_{nz_0}(h)$ is a set of integers in $N_{nz_0}(h)$. We have

$$\begin{aligned} \left\| \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{X}_t - \mathbf{X}_{z_n}) \mathbf{X}_{z_n}^\top \right\| &\leq \|\mathbf{X}_{z_n}\| \sum_{t=\lfloor (z_0-h)n \rfloor + 1}^{\lfloor (z_0+h)n \rfloor} K\left(\frac{t-nz_0}{nh}\right) \|\mathbf{X}_t - \mathbf{X}_{z_n}\| \\ &= O_p(\sqrt{n}) \times O(nh) \times O_p(\sqrt{nh}) = O_p(n^2 h^{3/2}). \end{aligned} \quad (2)$$

Similarly, we can show that

$$\left\| \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_{z_n} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \right\| = O_p(n^2 h^{3/2}), \quad \left\| \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{X}_t - \mathbf{X}_{z_n}) (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \right\| = O_p(n^2 h^2), \quad (3)$$

In view of (1), (2), and (3), we deduce that

$$\frac{1}{n^2 h} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^\top \Rightarrow \mathbf{K}_C(z_0) \mathbf{K}_C^\top(z_0).$$

A.2 Proof of Theorem 1

Recall that, we have the following decomposition

$$\begin{aligned} &(\hat{\mathbf{A}}_n(z_0) - \mathbf{A}(z_0)) \mathbf{Q}_n \mathbf{D}_n \\ &= \left[\left(\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{A}_t - \mathbf{A}(z_0)) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right] \left[\mathbf{D}_n^+ \mathbf{Q}_n^\top \left(\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right]^+ \\ &\quad + \left[\left(\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right] \left[\mathbf{D}_n^+ \mathbf{Q}_n^\top \left(\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \right]^+ \\ &:= \mathbf{B}_n(z_0) \Delta_n^+(z_0) + \Gamma_n(z_0) \Delta_n^+(z_0). \end{aligned}$$

We shall analyze $\mathbf{B}_n(z_0)$. For $\Delta_n(z_0)$ and $\Gamma_n(z_0)$, they are given in Lemma A1 and Lemma A2, respectively. Notice that

$$\begin{aligned} \mathbf{B}_n(z_0) &= \left(\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{A}_t - \mathbf{A}(z_0)) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \\ &= \left[\frac{1}{n\sqrt{h}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{A}_t - \mathbf{A}(z_0)) \mathbf{X}_t \mathbf{X}_t^\top \mathbf{q}_n \quad \frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{A}_t - \mathbf{A}(z_0)) \mathbf{X}_t \mathbf{X}_t^\top \mathbf{q}_n^\perp \right] \\ &:= [\mathbf{B}_n(z_0, 1) \quad \mathbf{B}_n(z_0, 2)]. \end{aligned}$$

Let us first consider $\mathbf{B}_n(z_0, 1)$. By a first-order Taylor expansion of $A(\cdot)$ around $z_0 \in (0, 1)$, we have

$$\frac{1}{n\sqrt{h}}\mathbf{B}_n(z_0, 1) = \nabla\mathbf{A}(z_0)\left(\frac{1}{n^2h}\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right)\left(\frac{t}{n}-z_0\right)\mathbf{X}_t\mathbf{X}_t^\top\right)\mathbf{q}_n + \text{s.o.},$$

where s.o. denotes higher-order negligible terms. Observe that

$$\frac{1}{nh}\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right)\left(\frac{t}{n}-z_0\right) = h\int_{-1}^1 uK(u)du + O(1/n) = O(1/n),$$

where $O(1/n)$ in the first line comes from the Riemann-sum approximation of an integral. Using similar arguments to establish Lemma 1 and Lemma A1, we can show that

$$\begin{aligned} & \frac{1}{n\sqrt{h}}\mathbf{B}_n(z_0, 1) \\ &= \nabla\mathbf{A}(z_0)\left(\frac{1}{nh}\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right)\left(\frac{t}{n}-z_0\right)\right)\left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}}\right)\left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}}\right)^\top \mathbf{q}_n \\ & \quad + \nabla\mathbf{A}(z_0) \cdot \frac{1}{n\sqrt{h}}\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right)\left(\frac{t}{n}-z_0\right)\left(\frac{\mathbf{X}_t-\mathbf{X}_{z_n}}{\sqrt{nh}}\right)\left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}}\right)^\top \mathbf{q}_n \\ & \quad + \nabla\mathbf{A}(z_0) \cdot \frac{1}{n\sqrt{h}}\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right)\left(\frac{t}{n}-z_0\right)\left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}}\right)\left(\frac{\mathbf{X}_t-\mathbf{X}_{z_n}}{\sqrt{nh}}\right)^\top \mathbf{q}_n \\ & \quad + \nabla\mathbf{A}(z_0) \cdot \frac{1}{n}\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right)\left(\frac{t}{n}-z_0\right)\left(\frac{\mathbf{X}_t-\mathbf{X}_{z_n}}{\sqrt{nh}}\right)\left(\frac{\mathbf{X}_t-\mathbf{X}_{z_n}}{\sqrt{nh}}\right)^\top \mathbf{q}_n + \text{s.o.} \\ &= O_p(1/n) + O_p(h^{3/2}) + O_p(h^{3/2}) + O_p(h^2) = O_p(h^{3/2}), \end{aligned}$$

which implies that $\|\mathbf{B}_n(z_0, 1)\| = O_p(nh^2)$. An entirely analogous argument shows that $\|\mathbf{B}_n(z_0, 2)\| = O_p(nh^{3/2})$. Then,

$$\begin{aligned} \left\|\left(\hat{\mathbf{A}}_n(z_0) - \mathbf{A}(z_0)\right)\mathbf{Q}_n\mathbf{D}_n - \mathbf{\Gamma}_n(z_0)\mathbf{\Delta}_n^+(z_0)\right\| &\leq \|\mathbf{B}_n(z_0)\| \|\mathbf{\Delta}_n^+(z_0)\| \\ &\leq (\|\mathbf{B}_n(z_0, 1)\| + \|\mathbf{B}_n(z_0, 2)\|) \|\mathbf{\Delta}_n^+(z_0)\| \\ &= O_p(nh^{3/2}). \end{aligned}$$

Since $\mathbf{\Delta}_n(z_0)$ and $\mathbf{\Gamma}_n(z_0)$ are both functions of the same localized partial-sum process of the innovation vector $\mathbf{u}_t$ together with the same random rotation based on X_{z_n} , their joint convergence follows directly by a straightforward application of the continuous mapping theorem to a vector collecting the relevant block components. Theorem 1 follows by combining Lemma A1 and Lemma A2.

A.3 Proof of Lemma A1

We follow the steps as in Phillips et al. (2017). Observe that

$$\begin{aligned}
& \mathbf{D}_n^+ \mathbf{Q}_n^\top \left(\sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{X}_t \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \\
&= \begin{bmatrix} \frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{q}_n^\top \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right) \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right)^\top \mathbf{q}_n & \frac{1}{nh^{3/2}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{q}_n^\top \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right) \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right)^\top \mathbf{q}_n^\perp \\ \frac{1}{nh^{3/2}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{q}_n^\perp)^\top \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right) \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right)^\top \mathbf{q}_n & \frac{1}{nh^2} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) (\mathbf{q}_n^\perp)^\top \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right) \left(\frac{\mathbf{X}_t}{\sqrt{n}}\right)^\top \mathbf{q}_n^\perp \end{bmatrix} \\
&:= \begin{bmatrix} \Delta_n(z_0, 1) & \Delta_n(z_0, 2) \\ \Delta_n(z_0, 2)^\top & \Delta_n(z_0, 3) \end{bmatrix}.
\end{aligned}$$

Let us first consider $\Delta_n(z_0, 1)$. Following the proof of Lemma 1, we can show that

$$\begin{aligned}
\Delta_n(z_0, 1) &= \frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{q}_n^\top \left(\frac{\mathbf{X}_t - \mathbf{X}_{z_n} + \mathbf{X}_{z_n}}{\sqrt{n}} \right) \left(\frac{\mathbf{X}_t - \mathbf{X}_{z_n} + \mathbf{X}_{z_n}}{\sqrt{n}} \right)^\top \mathbf{q}_n \\
&= \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right)^\top \mathbf{q}_n \left(\frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \right) + o_p(1) \\
&= \mathbf{q}_n^\top \mathbf{k}_n \mathbf{k}_n^\top \mathbf{q}_n + o_p(1) \\
&= \mathbf{k}_n^\top \mathbf{k}_n + o_p(1).
\end{aligned} \tag{4}$$

Now we consider $\Delta_n(z_0, 2)$. Recall that $z_n = \lfloor n(z_0 - h) \rfloor$, we have the decomposition

$$\mathbf{X}_t = \mathbf{X}_{z_n} + \sum_{j=z_n+1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj} + \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} + (\mathbf{R}_n^t - \mathbf{R}_n^{z_n}) \mathbf{X}_0,$$

where

$$\boldsymbol{\eta}_t := \sum_{j=z_n+1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj}, \quad \boldsymbol{\xi}_t := \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} + (\mathbf{R}_n^t - \mathbf{R}_n^{z_n}) \mathbf{X}_0.$$

Since by construction $(\mathbf{q}_n^\perp)^\top \mathbf{X}_{z_n} = 0$ a.s., we have

$$\begin{aligned}
\Lambda_n(z_0, 2) &= \frac{1}{nh^{3/2}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{q}_n^\top \left(\frac{\mathbf{X}_t}{\sqrt{n}} \right) \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp + o_p(1) \\
&= \frac{1}{nh^{3/2}} \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp \\
&\quad + \frac{1}{nh^{3/2}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{q}_n^\top \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right) \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp + o_p(1) \\
&:= \Lambda_n(z_0, 2, 1) + \Lambda_n(z_0, 2, 2) + o_p(1).
\end{aligned}$$

By Lemma B2(ii), we have $\sup_{t \in \bar{N}_{nz_0}(h)} \|\boldsymbol{\eta}_t + \boldsymbol{\xi}_t\| = O_p(\sqrt{nh})$, where $\bar{N}_{nz_0}(h)$ is a set of integers in $N_{nz_0}(h) = [\lfloor (z_0 - h)n \rfloor, \lfloor (z_0 + h)n \rfloor]$. This implies that

$$\begin{aligned} \|\Lambda_n(z_0, 2, 2)\| &\leq \frac{1}{nh^{3/2}} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left| \mathbf{q}_n^\top \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right) \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp \right| \\ &\leq \frac{1}{nh^{3/2}} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left\| \frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right\|^2 \\ &\leq \frac{1}{nh^{3/2}} \cdot \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \cdot \sup_{t \in \bar{N}_{nz_0}(h)} \left\| \frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}} \right\|^2 \\ &= \frac{1}{nh^{3/2}} \cdot O(nh) \cdot O_p(h) = O_p(\sqrt{h}). \end{aligned}$$

The dominating term is thus $\Lambda_n(z_0, 2, 1)$. We have

$$\begin{aligned} \Lambda_n(z_0, 2, 1) &= \frac{1}{nh^{3/2}} \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left(\frac{\boldsymbol{\xi}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp \\ &\quad + \frac{1}{nh^{3/2}} \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left(\frac{\boldsymbol{\eta}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp + o_p(1). \end{aligned}$$

Obviously

$$\begin{aligned} &\left\| \frac{1}{nh^{3/2}} \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left(\frac{\boldsymbol{\xi}_t}{\sqrt{n}} \right)^\top \mathbf{q}_n^\perp \right\| \\ &\leq \frac{1}{\sqrt{nh}} \cdot \left\| \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \right\| \cdot \left(\frac{1}{nh} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \|\boldsymbol{\xi}_t^\top \mathbf{q}_n^\perp\| \right) \\ &\leq \frac{1}{\sqrt{nh}} \cdot \left\| \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \right\| \cdot \sup_{t \in \bar{N}_{nz_0}(h)} \|\boldsymbol{\xi}_t\| \cdot \left(\frac{1}{nh} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \right) = o_p(1). \end{aligned}$$

Then, we have

$$\Delta_n(z_0, 2) = \frac{\sqrt{2}}{nh} \mathbf{q}_n^\top \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}} \right) \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left(\frac{\boldsymbol{\eta}_t}{\sqrt{2nh}} \right)^\top \mathbf{q}_n^\perp + o_p(1). \quad (5)$$

Finally, following the proof of (5) and the fact that $(\mathbf{q}_n^\perp)^\top \mathbf{X}_{z_n} = 0$ *a.s.*, we can show that

$$\begin{aligned}\Lambda_n(z_0, 3) &= \frac{1}{nh^2} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) (\mathbf{q}_n^\perp)^\top \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}}\right) \left(\frac{\boldsymbol{\eta}_t + \boldsymbol{\xi}_t}{\sqrt{n}}\right)^\top \mathbf{q}_n^\perp + o_p(1) \\ &= \frac{1}{nh^2} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) (\mathbf{q}_n^\perp)^\top \left(\frac{\boldsymbol{\eta}_t}{\sqrt{n}}\right) \left(\frac{\boldsymbol{\eta}_t}{\sqrt{n}}\right)^\top \mathbf{q}_n^\perp + o_p(1) \\ &= (\mathbf{q}_n^\perp)^\top \left[\frac{2}{nh} \sum_{t=1}^n K\left(\frac{t - nz_0}{nh}\right) \left(\frac{\boldsymbol{\eta}_t}{\sqrt{2nh}}\right) \left(\frac{\boldsymbol{\eta}_t}{\sqrt{2nh}}\right)^\top \right] \mathbf{q}_n^\perp + o_p(1).\end{aligned}\tag{6}$$

Using the invariance principle for linear processes under Assumption 3 and Lemma B2(i), there exist two independent Ornstein–Uhlenbeck processes $\mathbf{K}_C(\cdot)$ and $\mathbf{K}_C^*(\cdot)$ such that, for any fixed $z_0 \in (0, 1)$ and $0 < p \leq 1$,

$$\left(\begin{array}{c} \mathbf{k}_n \\ \frac{1}{\sqrt{2nh}} \sum_{j=z_n+1}^{z_n(p)} \mathbf{R}_n^{z_n(p)-j} \mathbf{u}_{xj} \end{array} \right) \Rightarrow \left(\begin{array}{c} \mathbf{k} \\ \mathbf{K}_C^*(p) \end{array} \right), \quad \mathbf{k}_n = \frac{\mathbf{X}_{z_n}}{\sqrt{n}}, \mathbf{k} = \mathbf{K}_C(z_0).$$

Since all block components in $\Delta_n(z_0)$ are functions of the same localized partial-sum process of X_t around z_0 and the same random rotation based on X_{z_n} , their joint convergence follows directly, which may be viewed as a straightforward application of the continuous mapping theorem to a vector of block components. Lemma A1 then follows by combining (4)-(6).

Now we show that $\Delta(z_0)$ is nonsingular *a.s.*. In view of Proposition 3.9.7. in Bernstein (2018), we need to verify that $\Delta(z_0, 1) > 0$ *a.s.* and $\Delta(z_0, 3) - \frac{1}{\Delta(z_0, 1)} \Delta^\top(z_0, 2) \Delta(z_0, 2)$ is positive definite *a.s.*. Given that $\mathbf{k} \equiv \mathbf{k}_{z_0} = \mathbf{K}_C(z_0)$ is Gaussian at fixed $z_0 \in (0, 1)$ with full rank covariance matrix $\Sigma_K(z_0) := \int_0^{z_0} e^{(z_0-s)\mathbf{C}} \boldsymbol{\Omega}_{xx} e^{(z_0-s)\mathbf{C}'} ds$, $\mathbb{P}(\mathbf{k} = \mathbf{0}) = 0$ so that $\mathbb{P}(\mathbf{k}^\top \mathbf{k} = 0) = 0$.

Next, given that $\int_{-1}^1 K(r) dr = 1$, we can write (by ignoring normalizing constant)

$$\begin{aligned}& \Delta(z_0, 3) - \frac{1}{\Delta(z_0, 1)} \Delta^\top(z_0, 2) \Delta(z_0, 2) \\ &= (\mathbf{q}^\perp)^\top \left[\int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) dr \right. \\ &\quad \left. - \left(\int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) dr \right) \left(\int_{-1}^1 K(r) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) dr \right) \right] \mathbf{q}^\perp \\ &= (\mathbf{q}^\perp)^\top \left[\int_{-1}^1 K(r) \left(\mathbf{K}_C^* \left(\frac{r+1}{2} \right) - \int_{-1}^1 K(r) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) dr \right) \right. \\ &\quad \left. \cdot \left(\mathbf{K}_C^* \left(\frac{r+1}{2} \right) - \int_{-1}^1 K(r) \mathbf{K}_C^{*\top} \left(\frac{r+1}{2} \right) dr \right)^\top \right] \mathbf{q}^\perp.\end{aligned}$$

Take any $\mathbf{x} \in \mathbb{R}^{k-1} \neq \mathbf{0}$ and let $\omega := \mathbf{q}^\perp \mathbf{x}$. We can write the above as

$$\begin{aligned} & \mathbf{x}^\top (\mathbf{q}^\perp)^\top \left[\int_{-1}^1 K(r) \left(\mathbf{K}_C^* \left(\frac{r+1}{2} \right) - \int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) dr \right) \right. \\ & \quad \left. \cdot \left(\mathbf{K}_C^* \left(\frac{r+1}{2} \right) - \int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) dr \right)^\top \right] \mathbf{q}^\perp \\ &= \int_{-1}^1 K(r) \left(\omega^\top \left(\mathbf{K}_C^* \left(\frac{r+1}{2} \right) - \int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) dr \right) \right)^2 dr \geq 0, \quad a.s. \end{aligned}$$

What remains is to show that

$$\mathbb{P} \left\{ \int_{-1}^1 K(r) \left(\omega^\top \left(\mathbf{K}_C^* \left(\frac{r+1}{2} \right) - \int_{-1}^1 K(r) \mathbf{K}_C^* \left(\frac{r+1}{2} \right) dr \right) \right)^2 dr = 0 \right\} = 0. \quad (7)$$

Since Assumption 1 implies that $K(r) \geq 0$ and is positive on a set of nonzero Lebesgue measure in $[-1, 1]$, on the event that (7) holds we must have

$$\omega^\top \mathbf{K}_C^* \left(\frac{r+1}{2} \right) = \omega^\top \int_{-1}^1 K(s) \mathbf{K}_C^* \left(\frac{s+1}{2} \right) ds, \quad a.s.$$

for all $r \in [-1, 1]$ with $K(r) > 0$ except on a set of Lebesgue measure zero. This can only happen if for almost every realization, the sample path $r \mapsto \omega^\top \mathbf{K}_C^* \left(\frac{r+1}{2} \right)$ is constant in r , which has probability zero since $\mathbf{K}_C^* \left(\frac{r+1}{2} \right)$ is a nondegenerate continuous Gaussian process.

A.4 Proof of Lemma A2

We have

$$\begin{aligned} \Gamma_n(z_0) &= \left(\sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) \mathbf{u}_{0t} \mathbf{X}_t^\top \right) \mathbf{Q}_n \mathbf{D}_n^+ \\ &= \left[\frac{1}{n\sqrt{h}} \sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) \mathbf{u}_{0t} \mathbf{X}_t^\top \mathbf{q}_n \quad \frac{1}{nh} \sum_{t=1}^n K \left(\frac{t - nz_0}{nh} \right) \mathbf{u}_{0t} \mathbf{X}_t^\top \mathbf{q}_n^\perp \right] \\ &:= \begin{bmatrix} \Gamma_n(z_0, 1) & \Gamma_n(z_0, 2) \end{bmatrix}. \end{aligned}$$

For $\Gamma_n(z_0, 1)$, notice that

$$\begin{aligned}
\Gamma_n(z_0, 1) &= \frac{1}{n\sqrt{h}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n} + \mathbf{X}_{z_n})^\top \mathbf{q}_n \\
&= \frac{1}{\sqrt{2nh}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \cdot \sqrt{2} \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}}\right)^\top \mathbf{q}_n + \frac{1}{n\sqrt{h}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \mathbf{q}_n \\
&:= \Gamma_n(z_0, 1, 1) + \Gamma_n(z_0, 1, 2).
\end{aligned}$$

Using Lemma B1, we obtain

$$\begin{aligned}
\|\Gamma_n(z_0, 1, 2)\| &\leq \frac{1}{n\sqrt{h}} \cdot \left\| \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \right\| \cdot \|\mathbf{q}_n\| \\
&= \frac{1}{n\sqrt{h}} \cdot O_p(nh) = O_p(\sqrt{h}),
\end{aligned}$$

so that the dominating term is $\Gamma_n(z_0, 1, 1)$. For $\Gamma_n(z_0, 2)$, given that $(\mathbf{q}_n^\perp)^\top \mathbf{X}_{z_n} = 0$ a.s., we have

$$\begin{aligned}
\Gamma_n(z_0, 2) &= \frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n} + \mathbf{X}_{z_n})^\top \mathbf{q}_n^\perp \\
&= \frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \mathbf{q}_n^\perp + o_p(1) := \Gamma_n(z_0, 2, 1) + o_p(1).
\end{aligned}$$

Then, using the weak convergence result of

$$\left(\begin{array}{c} \frac{1}{\sqrt{2nh}} \sum_{j=z_n+1}^{z_n(p)} \mathbf{u}_{0j} \\ \frac{1}{\sqrt{2nh}} \sum_{j=z_n+1}^{z_n(p)} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \end{array} \right) \Rightarrow \left(\begin{array}{c} \mathbf{B}_0(p) \\ \mathbf{K}_C(p) \end{array} \right),$$

where $z_n(p) = z_n + \lfloor 2pn \rfloor + 1$ for $0 < p \leq 1$, Lemma B1, and the continuous mapping theorem, we can show that

$$\begin{aligned}
&\left[\Gamma_n(z_0, 1, 1) \quad \Gamma_n(z_0, 2, 1) \right] \\
&= \left[\frac{1}{\sqrt{2nh}} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \cdot \sqrt{2} \left(\frac{\mathbf{X}_{z_n}}{\sqrt{n}}\right)^\top \mathbf{q}_n \quad \frac{1}{nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \mathbf{q}_n^\perp \right] \\
&\Rightarrow \left[\sqrt{2} \|\mathbf{k}\| \int_{-1}^1 K(r) d\mathbf{B}_0^*\left(\frac{r+1}{2}\right) \quad \left[\int_{-1}^1 K(r) d\mathbf{B}_0^*\left(\frac{r+1}{2}\right) \mathbf{K}_C^{*\top}\left(\frac{r+1}{2}\right) + \frac{1}{2} (\Sigma_{0x} + \Lambda_{0x}) \right] \mathbf{q}^\perp \right].
\end{aligned}$$

B Auxiliary results

Lemma B1. *If Assumptions 1-3 are satisfied. Then, for any fixed $z_0 \in (0, 1)$,*

$$\frac{1}{2nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \Rightarrow \int_{-1}^1 d\mathbf{B}_0^*\left(\frac{r+1}{2}\right) K(r) \mathbf{K}_C^{*\top}\left(\frac{r+1}{2}\right) + \frac{1}{2} [\Sigma_{0x} + \Lambda_{0x}],$$

where $z_n = \lfloor n(z_0 - h) \rfloor$ and $(\mathbf{K}_C^{*\top}(r), \mathbf{B}_0^{*\top}(r))^\top$ is an independent copy of $(\mathbf{K}_C^\top(r), \mathbf{B}_0^\top(r))^\top$. $\mathbf{B}_0(r)$ is a k -vector Brownian motion with covariance matrix Ω_{00} .

Proof. Let $\bar{N}_{nz_0}(h)$ be a set of integers in $N_{nz_0}(h) = [\lfloor (z_0 - h)n \rfloor, \lfloor (z_0 + h)n \rfloor]$ and $z_n = \lfloor n(z_0 - h) \rfloor$. We have

$$\begin{aligned} & \frac{1}{2nh} \sum_{t=1}^n K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} (\mathbf{X}_t - \mathbf{X}_{z_n})^\top \\ &= \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \left\{ \sum_{j=z_n+1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj} + \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} + (\mathbf{R}_n^t - \mathbf{R}_n^{z_n}) \mathbf{X}_0 \right\}^\top \\ &= \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \left\{ \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\}^\top + \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \mathbf{u}_{xt}^\top \\ &\quad + \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}^+(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \left\{ \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} \right\}^\top \\ &\quad + \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \{(\mathbf{R}_n^t - \mathbf{R}_n^{z_n}) \mathbf{X}_0\}^\top \\ &:= \sum_{i=1}^4 \mathbf{S}_{in}. \end{aligned}$$

We first show that $\|\mathbf{S}_{3n}\| = o_p(1)$. To see this, note that

$$\|\mathbf{S}_{3n}\| \leq \sup_{t \in \bar{N}_{nz_0}^+(h)} \left\| \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} \right\| \left\| \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}^+(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \right\|.$$

Following the proofs of Lemma B2(ii) and the central limit theorem (C.L.T.) for weighted linear processes (Abadir et al., 2014; Yan et al., 2025), we have $\sup_{t \in \bar{N}_{nz_0}(h)} \left\| \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} \right\| = o_p(\sqrt{nh})$ and $\left\| \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t-nz_0}{nh}\right) \mathbf{u}_{0t} \right\| = O_p\left(\frac{1}{\sqrt{nh}}\right)$. This implies that $\|\mathbf{S}_{3n}\| = o_p(1)$. Similar arguments also yields $\|\mathbf{S}_{4n}\| = o_p(1)$.

We now move on to $\mathbf{S}_{1n}$ and $\mathbf{S}_{2n}$. $\mathbf{S}_{2n}$ is straightforward, since by the weak law of large numbers

(W.L.L.N.) for weighted linear processes (Abadir et al., 2014; Yan et al., 2025), we have

$$2\mathbf{S}_{2n} \xrightarrow{p} \Sigma_{0x}.$$

What remains are $\mathbf{S}_{1n}$. BN decomposition on $\mathbf{u}_{0t}$ gives

$$\begin{aligned} \mathbf{S}_{1n} &= \Phi_0(1) \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \varepsilon_t \left\{ \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\}^\top \\ &\quad - \frac{1}{2nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \Delta \tilde{\mathbf{u}}_{0t} \left\{ \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\}^\top \\ &:= \mathbf{S}_{1n}^{(1)} + \mathbf{S}_{1n}^{(2)}. \end{aligned}$$

For $\mathbf{S}_{1n}^{(1)}$, let

$$\mathbf{U}_n(r) := \Phi_0(1) \frac{1}{\sqrt{2nh}} \sum_{j=z_n+1}^{z_n + \lfloor 2nhr \rfloor + 1} \varepsilon_j, \quad \mathbf{V}_n(r) := \frac{1}{\sqrt{2nh}} \sum_{j=z_n+1}^{z_n + \lfloor 2nhr \rfloor + 1} \mathbf{R}_n^{z_n + \lfloor 2nhr \rfloor - j} \mathbf{u}_{xj}, \quad r \in [0, 1].$$

We note that

$$\begin{aligned} \mathbf{S}_{1n}^{(1)} &= \sum_{t \in \bar{N}_{nz_0}(h)} \left(\frac{1}{\sqrt{2nh}} \Phi_0(1) \varepsilon_t \right) \left\{ \frac{1}{\sqrt{2nh}} K\left(\frac{t - nz_0}{nh}\right) \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\}^\top \\ &= \int_0^1 K(r) d\mathbf{U}_n(r) \mathbf{V}_n^\top(r) + o_p(1). \end{aligned}$$

Using the weak convergence result of

$$\begin{pmatrix} \mathbf{U}_n(r) \\ \mathbf{V}_n(r) \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{B}_0(r) \\ \mathbf{K}_C(r) \end{pmatrix}, \quad r \in [0, 1],$$

we obtain

$$\mathbf{S}_{1n}^{(1)} \Rightarrow \int_{-1}^1 K(r) d\mathbf{B}_0^*((r+1)/2) \mathbf{K}_C^{*\top}((r+1)/2),$$

where $(\mathbf{K}_C^{*\top}(r), \mathbf{B}_0^{*\top}(r))^\top$ is an independent copy of $(\mathbf{K}_C^\top(r), \mathbf{B}_0^\top(r))^\top$. Moving on to $\mathbf{S}_{1n}^{(2)}$, summation by parts

yields

$$\begin{aligned}
2\mathbf{S}_{1n}^{(2)} &= \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \left\{ \sum_{j=z_n+1}^t \mathbf{R}_n^{t+1-j} \mathbf{u}_{xj} - \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\}^\top \\
&+ \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} \left\{ K\left(\frac{t - nz_0}{nh}\right) - K\left(\frac{t - 1 - nz_0}{nh}\right) \right\} \tilde{\mathbf{u}}_{0,t-1} \left(\sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right)^\top \\
&= \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \left(\sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right)^\top (\mathbf{R}_n - \mathbf{I}_k)^\top + \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \mathbf{u}_{xt}^\top \mathbf{R}_n^\top \\
&+ \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} \left\{ K\left(\frac{t - nz_0}{nh}\right) - K\left(\frac{t - 1 - nz_0}{nh}\right) \right\} \tilde{\mathbf{u}}_{0,t-1} \left(\sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right)^\top.
\end{aligned}$$

Note that

$$\begin{aligned}
&\left\| \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}^+(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \left(\sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right)^\top (\mathbf{R}_n - \mathbf{I}_k)^\top \right\| \\
&\leq \|\mathbf{R}_n - \mathbf{I}_k\| \sup_{t \in \bar{N}_{nz_0}(h)} \left\| \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\| \left\| \left(\frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \|\tilde{\mathbf{u}}_{0t}\| \right) \right\| \\
&= O(1/n) O_p(\sqrt{nh}) = o_p(1),
\end{aligned}$$

which follows from the fact that by Lemma B2(i) we have $\sup_{t \in \bar{N}_{nz_0}(h)} \left\| \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\| = O_p(\sqrt{nh})$ and $\|\tilde{\mathbf{u}}_{0t}\| = O_p(1)$ holds uniformly over $t \in \bar{N}_{nz_0}(h)$. We also have

$$\begin{aligned}
&\left\| \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} \left\{ K\left(\frac{t - nz_0}{nh}\right) - K\left(\frac{t - 1 - nz_0}{nh}\right) \right\} \tilde{\mathbf{u}}_{0,t-1} \left(\sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right)^\top \right\| \\
&\leq \sup_{t \in \bar{N}_{nz_0}(h)} \left\| \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\| \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} \left\| \left\{ K\left(\frac{t - nz_0}{nh}\right) - K\left(\frac{t - 1 - nz_0}{nh}\right) \right\} \tilde{\mathbf{u}}_{0,t-1} \right\| \\
&\leq \sup_{t \in \bar{N}_{nz_0}(h)} \left\| \sum_{j=z_n+1}^{t-1} \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\| \left(\frac{O(1)}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} (1/nh) \|\tilde{\mathbf{u}}_{0t}\| \right) \\
&= O_p(\sqrt{nh}) O(1/nh) = o_p(1),
\end{aligned}$$

where we have used the fact $K(\cdot)$ is Lipschitz-continuous on $[-1, 1]$. Finally, by the weak law of large

numbers (W.L.L.N.) for weighted linear processes (Abadir et al., 2014; Yan et al., 2025), we have

$$\frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \mathbf{u}_{xt}^\top \xrightarrow{p} \Lambda_{0x}.$$

Given that

$$\begin{aligned} & \left\| \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \mathbf{u}_{xt}^\top \mathbf{R}_n^\top - \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \mathbf{u}_{xt}^\top \right\| \\ & \leq \|\mathbf{R}_n - \mathbf{I}_k\| \left\| \frac{1}{nh} \sum_{t \in \bar{N}_{nz_0}(h)} K\left(\frac{t - nz_0}{nh}\right) \tilde{\mathbf{u}}_{0t} \mathbf{u}_{xt}^\top \right\| = O_p(1/n) = o_p(1), \end{aligned}$$

together with Slutsky's theorem, leads to

$$\mathbf{S}_{1n} \Rightarrow \int_{-1}^1 K(r) d\mathbf{B}_0^*((r+1)/2) \mathbf{K}_C^{*\top}((r+1)/2) + \frac{1}{2} \Lambda_{0x}.$$

The proofs are completed by combining the results above. $\square$

Lemma B2. Consider the nearly integrated system as in Section 3. Let $\mathbf{u}_t = (\mathbf{u}_{0t}^\top, \mathbf{u}_{xt}^\top)^\top$ be defined as in (6). Under Assumptions 1, 2, and 3, we have, as $n \rightarrow \infty$:

- (i) $n^{-1/2} \mathbf{X}_{\lfloor nr \rfloor} \Rightarrow \mathbf{K}_C(r)$, $0 < r \leq 1$;
- (ii) $\sup_{t \in \bar{N}_{nz_0}(h)} \left\| \frac{\mathbf{X}_t - \mathbf{X}_{z_n}}{\sqrt{2nh}} \right\| = O_p(1)$, where $\bar{N}_{nz_0}(h)$ is a set of integers in $N_{nz_0}(h) = [\lfloor (z_0 - h)n \rfloor, \lfloor (z_0 + h)n \rfloor]$ and $z_n = \lfloor n(z_0 - h) \rfloor$, $z_0 \in (0, 1)$.

Proof. (i) is almost identical to Lemma 3.1(a) in Phillips (1988). However, since we have a different formulation for $\mathbf{R}_n$, the proofs are recollected here for completeness. Let $\mathbf{S}_t^x = \sum_{j=1}^t \mathbf{u}_{xj}$ be the vector of partial sums. We construct

$$\mathbf{Z}_n(r) = n^{-1/2} \mathbf{S}_{\lfloor nr \rfloor}^x = n^{-1/2} \mathbf{S}_{j-1}^x, \quad \frac{j-1}{n} \leq r < \frac{j}{n},$$

where $j = 1, 2, \dots, n$. As shown in Phillips (1987), $\mathbf{Z}_n(r)$ obeys a functional central limit theorem:

$$\mathbf{Z}_n(r) \Rightarrow \mathbf{B}_X(r), \tag{8}$$

where $\mathbf{B}_X(r)$ is a k -vector Brownian motion with covariance matrix $\boldsymbol{\Omega}_{xx}$.

Based on the recursion

$$\mathbf{X}_t = \sum_{j=1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj} + \mathbf{R}_n^t \mathbf{X}_0,$$

we have

$$\begin{aligned}
n^{-1/2} \mathbf{X}_{[nr]} &= \sum_{j=1}^{[nr]} \left(\mathbf{I}_k + \frac{\mathbf{C}}{n} \right)^{[nr]-j} \int_{\frac{j-1}{n}}^{\frac{j}{n}} d\mathbf{Z}_n(s) + O_p(n^{-1/2}) \\
&= \sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} \left(\mathbf{I}_k + \frac{\mathbf{C}}{n} \right)^{[nr]-j} d\mathbf{Z}_n(s) + O_p(n^{-1/2}) \\
&= \sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} \left[\left(\mathbf{I}_k + \frac{\mathbf{C}}{n} \right)^{[nr]-j} - e^{(r-s)\mathbf{C}} \right] d\mathbf{Z}_n(s) \\
&\quad + \sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} e^{(r-s)\mathbf{C}} d\mathbf{Z}_n(s) + O_p(n^{-1/2}) \\
&\leq \mathbf{T}_n \sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} d\mathbf{Z}_n(s) + \sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} e^{(r-s)\mathbf{C}} d\mathbf{Z}_n(s) + O_p(n^{-1/2}),
\end{aligned} \tag{9}$$

where

$$\mathbf{T}_n := \max_{1 \leq j \leq n} \left\{ \sup_{\frac{j-1}{n} \leq s < \frac{j}{n}} \left\| \left(\mathbf{I}_k + \frac{\mathbf{C}}{n} \right)^{[nr]-j} - e^{(r-s)\mathbf{C}} \right\| \right\}.$$

Notice that

$$\begin{aligned}
\sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} e^{(r-s)\mathbf{C}} d\mathbf{Z}_n(s) &= \int_0^{[nr]/n} e^{(r-s)\mathbf{C}} d\mathbf{Z}_n(s) \\
&= \int_0^r e^{(r-s)\mathbf{C}} d\mathbf{Z}_n(s) + O_p(1/n) \\
&= \mathbf{Z}_n(r) + \mathbf{C} \int_0^r e^{(r-s)\mathbf{C}} \mathbf{Z}_n(s) ds + O_p(1/n).
\end{aligned}$$

By (8) and the continuous mapping theorem (C.M.T.), we have

$$\mathbf{Z}_n(r) + \mathbf{C} \int_0^r e^{(r-s)\mathbf{C}} \mathbf{Z}_n(s) ds \Rightarrow \mathbf{B}_X(r) + \mathbf{C} \int_0^r e^{(r-s)\mathbf{C}} \mathbf{B}_X(s) ds.$$

This implies that $\sum_{j=1}^{[nr]} \int_{\frac{j-1}{n}}^{\frac{j}{n}} e^{(r-s)\mathbf{C}} d\mathbf{Z}_n(s) \Rightarrow \mathbf{K}_C(r)$, where $\mathbf{K}_C(r) = \mathbf{B}_X(r) + \mathbf{C} \int_0^r e^{(r-s)\mathbf{C}} \mathbf{B}_X(s) ds$. What remains is to show that $\mathbf{T}_n = o_p(1)$. Observe that both $\mathbf{I}_k + \frac{\mathbf{C}}{n}$ and $e^{(r-s)\mathbf{C}}$ are diagonal matrix. It is suffice to verify that, for $i = 1, 2, \dots, k$,

$$\max_{1 \leq j \leq n} \left\{ \sup_{\frac{j-1}{n} \leq s < \frac{j}{n}} \left| \left(1 + \frac{c_i}{n} \right)^{[nr]-j} - e^{c_i(r-s)} \right| \right\} = o(1).$$

We follow closely the steps in the proofs of Lemma 2.2 in Chan and Wei (1987). Define $d_n^i = -\frac{n}{c_i} \log \left(1 + \frac{c_i}{n} \right)$.

Clearly, we have $d_n^i \rightarrow -1$ as $n \rightarrow \infty$. For $\frac{j-1}{n} \leq s < \frac{j}{n}$, $j = 1, 2, \dots, n$, observe that

$$\begin{aligned} \left| \left(1 + \frac{c_i}{n}\right)^{\lfloor nr \rfloor - j} - e^{c_i(r-s)} \right| &= e^{c_i(r-\frac{j}{n})} \left| e^{(d_n^i+1)c_i(\frac{j}{n}-r)} - e^{c_i(\frac{j}{n}-s)} \right| \\ &\leq e^{|c_i|} \left\{ \left| e^{(d_n^i+1)c_i(\frac{j}{n}-r)} - 1 \right| + \left| 1 - e^{c_i(\frac{j}{n}-s)} \right| \right\}. \end{aligned}$$

The proof is completed by noticing that

$$\max_{1 \leq j \leq n} \left| (d_n^i + 1)c_i\left(\frac{j}{n} - r\right) \right| \leq |c_i| |d_n^i + 1| \rightarrow 0,$$

and

$$\max_{1 \leq j \leq n} \sup_{\frac{j-1}{n} \leq s < \frac{j}{n}} \left| 1 - e^{c_i(\frac{j}{n}-s)} \right| \leq \max_{1 \leq j \leq n} \{e^{c_i/n} - 1, 1 - e^{c_i/n}\} \rightarrow 0,$$

as $n \rightarrow \infty$.

For (ii), we first notice that

$$\begin{aligned} \frac{1}{\sqrt{2[nh]}} (\mathbf{X}_t - \mathbf{X}_{z_n}) &= \frac{1}{\sqrt{2[nh]}} \sum_{j=z_n+1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj} + \frac{1}{\sqrt{2[nh]}} \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} \\ &\quad + \frac{1}{\sqrt{2[nh]}} (\mathbf{R}_n^t - \mathbf{R}_n^{z_n}) \mathbf{X}_0. \end{aligned}$$

Define the localized partial-sum process

$$\mathbf{S}_n(r) := \frac{1}{\sqrt{2[nh]}} \sum_{j=z_n+1}^{z_n+\lfloor 2[nh]r \rfloor} \mathbf{R}_n^{z_n+\lfloor 2[nh]r \rfloor-j} \mathbf{u}_{xj}, \quad r \in [0, 1].$$

Following the same arguments as in part (i), we have $\mathbf{S}_n(\cdot) \Rightarrow \mathbf{K}_C(\cdot)$. Since $\mathbf{K}_C(\cdot)$ has continuous sample paths on the compact set $[0, 1]$, the mapping $f \mapsto \sup_{r \in [0, 1]} \|f(r)\|$ is continuous. Hence, by C.M.T.,

$$\sup_{t \in \bar{N}_{nz_0}(h)} \left\| \frac{1}{\sqrt{2[nh]}} \sum_{j=z_n+1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\| \Rightarrow \sup_{0 < r \leq 1} \|\mathbf{K}_C(r)\|.$$

This implies that

$$\sup_{t \in \bar{N}_{nz_0}(h)} \left\| \frac{1}{\sqrt{2[nh]}} \sum_{j=z_n+1}^t \mathbf{R}_n^{t-j} \mathbf{u}_{xj} \right\| = O_p(1).$$

In addition, using the relation

$$\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j} = \mathbf{R}_n^{z_n-j} (\mathbf{R}_n^{t-z_n} - \mathbf{I}_k), \quad 1 \leq j \leq z_n, \quad (10)$$

we have

$$\sup_{t \in \bar{N}_{nz_0}(h)} \max_{1 \leq j \leq z_n} \|\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}\| \leq O(1) \sup_{|m| \leq nh} \|\mathbf{R}_n^m - \mathbf{I}_k\| = O(h), \quad (11)$$

which follows from the fact that $\sup_{0 \leq m \leq n} \|\mathbf{R}_n^m\| = O(1)$ and $\sup_{|m| \leq nh} \|\mathbf{R}_n^m - \mathbf{I}_k\| = O(h)$. Moreover, $\|\sum_{j=1}^{z_n} \mathbf{u}_{xj}\| = O_p(\sqrt{n})$, so $\left\| \frac{1}{\sqrt{2[nh]}} \sum_{j=1}^{z_n} \mathbf{u}_{xj} \right\| = O_p(1/\sqrt{h})$. Therefore,

$$\left\| \frac{1}{\sqrt{2[nh]}} \sum_{j=1}^{z_n} (\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}) \mathbf{u}_{xj} \right\| \leq \max_{1 \leq j \leq z_n} \|\mathbf{R}_n^{t-j} - \mathbf{R}_n^{z_n-j}\| \left\| \frac{1}{\sqrt{2[nh]}} \sum_{j=1}^{z_n} \mathbf{u}_{xj} \right\| = o_p(1),$$

uniformly over $t \in \bar{N}_{nz_0}(h)$. We also have

$$\left\| \frac{1}{\sqrt{2[nh]}} (\mathbf{R}_n^t - \mathbf{R}_n^{z_n}) \mathbf{X}_0 \right\| \leq \frac{1}{\sqrt{2[nh]}} \|\mathbf{R}_n^t - \mathbf{R}_n^{z_n}\| \|\mathbf{X}_0\| = o_p(1)$$

This completes the proof of (ii). □

C Additional simulation results

We present some additional simulation results based on Case 4: $\rho = 0.5, \rho_1 = -0.5, \rho_2 = 0.5, \lambda = 0.5$. The D.G.P. is exactly the same as described in Section 4. However, instead of using a pre-specified bandwidth h , we consider to select h based on the cross-validation method. In particular, we set $h = c \cdot n^{-2/3}$ with c ranging from 0.05 to 2 with step size 0.05. Then, c is selected by minimizing

$$\text{CV}_n(h) = \frac{1}{n} \sum_{t=1}^n \left(y_t - \widehat{\mathbf{A}}_{-t}^\top \left(\frac{t}{n} |h \right) \mathbf{X}_t \right)^2, \quad (12)$$

where $\widehat{\mathbf{A}}_{-t} \left(\frac{t}{n} |h \right)$ is the leave-one-out estimate

$$\widehat{\mathbf{A}}_{-t} \left(\frac{t}{n} |h \right) = \left[\sum_{s=1, s \neq t}^n K \left(\frac{s - nz_0}{nh} \right) \mathbf{y}_s \mathbf{X}_s^\top \right] \left[\sum_{s=1, s \neq t}^n K \left(\frac{s - nz_0}{nh} \right) \mathbf{X}_s \mathbf{X}_s^\top \right]^+.$$

After obtaining the optimal bandwidth $\hat{h}_{CV}$, we use it to calculate $\widehat{\mathbf{A}}_n(t/n)$ over $t = 1, 2, \dots, n$.

The plots shown in Figure 1 and Figure 2 are based on the transformed quantities

$$\begin{aligned} \tilde{a}_{1n}(z_0) &= q_{1n}(z_0) [\hat{a}_{1n}(z_0) - a_1(z_0)] + q_{2n}(z_0) [\hat{a}_{2n}(z_0) - a_2(z_0)], \\ \tilde{a}_{2n}(z_0) &= p_{1n}(z_0) [\hat{a}_{1n}(z_0) - a_1(z_0)] + p_{2n}(z_0) [\hat{a}_{2n}(z_0) - a_2(z_0)], \end{aligned}$$

where $\hat{\mathbf{A}}_t = (\hat{a}_{1n}(z_0), \hat{a}_{2n}(z_0))^T$ and the exact definition of $(q_{1n}(z_0), q_{2n}(z_0))^T$ and $(p_{1n}(z_0), p_{2n}(z_0))^T$ can be found in Section 4. These results are based on 500 replications, averaged at each t/n , where $n = 500$. Overall, these plots clearly show that fluctuations of $\tilde{a}_{2n}(z_0)$ are much larger than $\tilde{a}_{1n}(z_0)$, further confirming the limit theory that the convergence rate of $\tilde{a}_{1n}(z_0)$ is faster, so that the variability is also smaller.

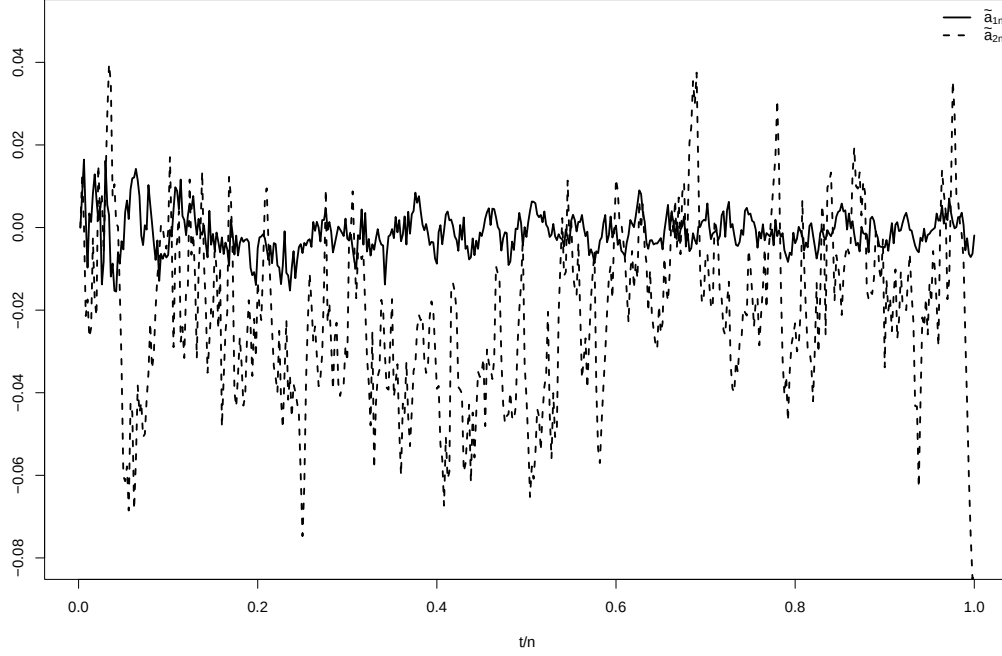


Figure 1: Plots of $\tilde{a}_{1n}(t/n)$ and $\tilde{a}_{2n}(t/n)$ under M1.

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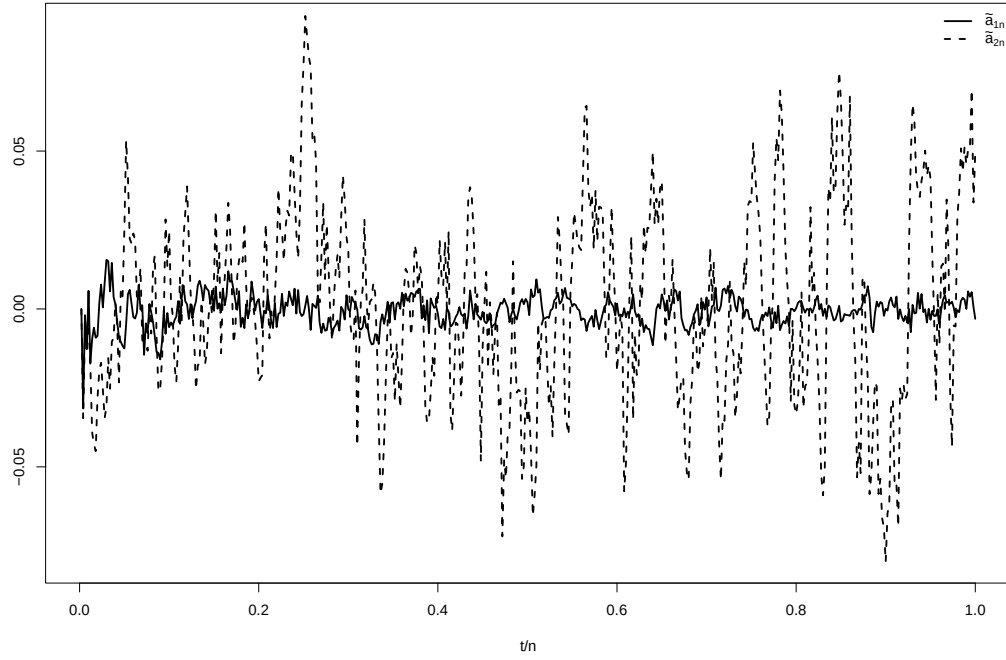


Figure 2: Plots of $\tilde{a}_{1n}(t/n)$ and $\tilde{a}_{2n}(t/n)$ under M2.

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